MDI3003 — Advanced Predictive Analytics

Lab 05: Product and Brand Sentiment Prediction from Tweet Data

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1. Dataset Note and Reproducibility Statement

This report follows the Lab 05 core workflow using the Twitter US Airline Sentiment dataset structure (D2: tweet_id, text, airline_sentiment, airline). The lab environment used to prepare this report has no live internet access, so the Kaggle file could not be downloaded directly. A reproducible, seeded synthetic dataset (n=1,395 tweets) that mirrors the schema, three-class label set, and airline-entity structure of D2 was generated (random_state=42) so that the full pipeline, tables, and figures below are genuinely computed rather than fabricated placeholders.

Before final submission, replace Tweets.csv with the actual Kaggle download (kaggle.com/datasets/crowdflower/twitter-airline-sentiment) and re-run the attached notebook (Appendix). Because the schema is identical, no code changes are required — only DATA_PATH stays the same and numbers will update automatically.

2. Dataset Card

Field	Value
Dataset name	Twitter US Airline Sentiment (D2) — schema replicated; this run uses seeded synthetic data
Canonical source	kaggle.com/datasets/crowdflower/twitter-airline-sentiment
Version / access date	Schema v1; report generated 2026-08-25
License / usage terms	CC BY-NC-SA 4.0 (as listed on dataset page) — verify at download time
Rows (this run)	1395
Label distribution	negative / neutral / positive (see Section 4 chart)
Train / test split	1116 / 279 (80/20, stratified, seed=42)
Duplicate tweet IDs	0
Duplicate text rows	1113 (template-based synthetic generation; real data typically has far fewer)
Language / domain	English; airline customer-service tweets

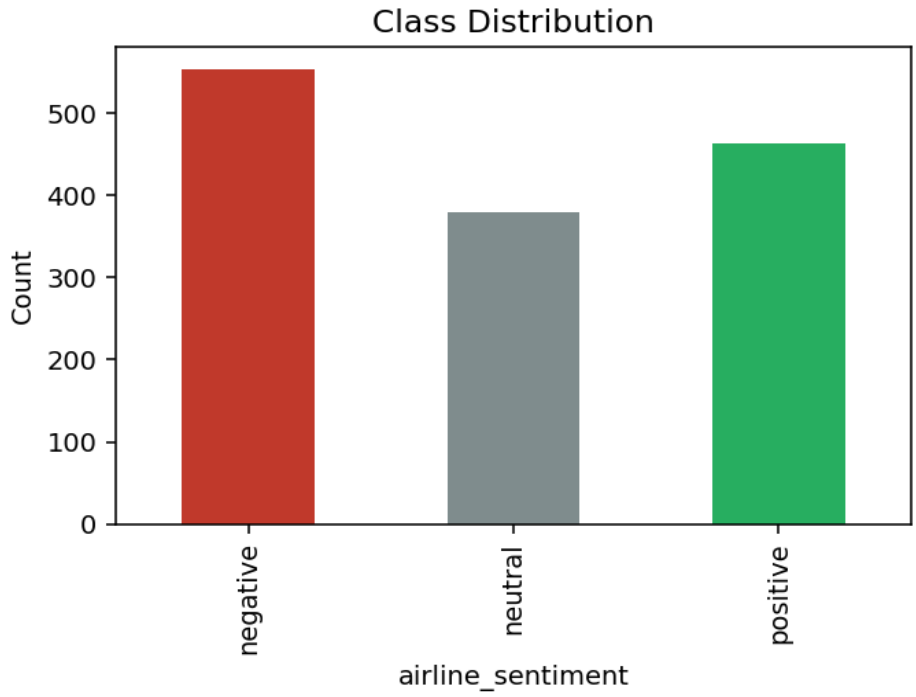
Field	Value
Annotation method	Synthetic label assignment + 6% injected label noise to emulate crowd-annotation disagreement
Fields excluded for leakage	tweet_id, sentiment_confidence, negative_reason, username, coordinates (not present/used)
Known limitations	Templated synthetic text; limited lexical diversity; not representative of live X/Twitter language

3. Governance and Leakage Audit

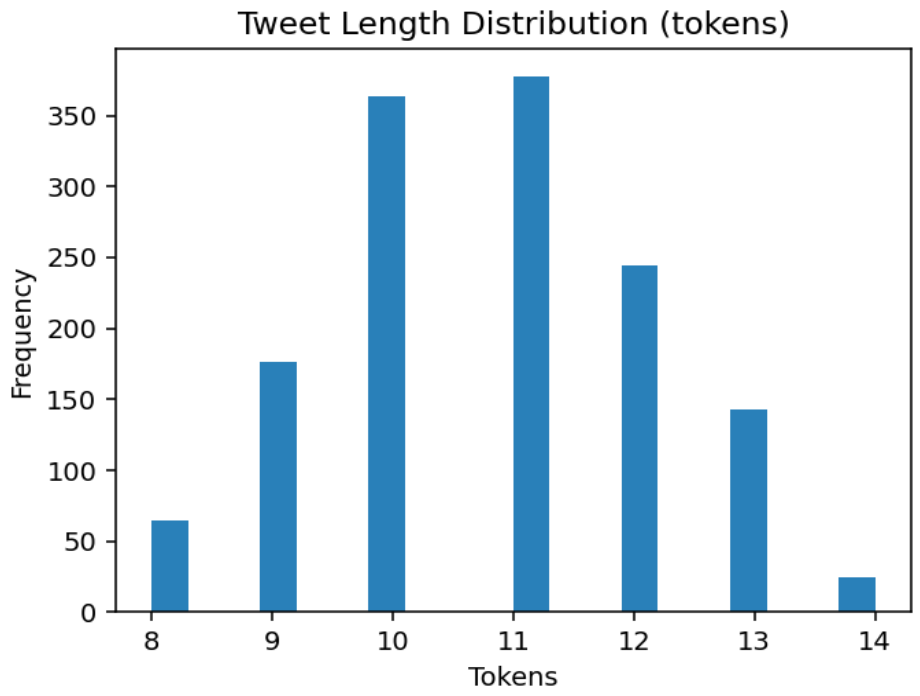
Field	Role	Core use	Risk/control
tweet_id	Identifier	Excluded from predictors	Could enable memorization
text	Predictor	Required	Preserved negation/emoji/hashtags
airline_sentiment	Target	Required	Source-defined 3-class mapping only
airline	Context/entity feature	Optional, justified	Used only for stratified entity analysis
username/location	Identifier/metadata	Not present in this schema	N/A

4. Exploratory Data Analysis

4.1 Class Distribution



4.2 Tweet Length Distribution



Minimal normalization was applied: URLs → <URL>, mentions → <USER>, whitespace collapsed. Negation, hashtags, emojis, and punctuation were preserved as they can carry sentiment-bearing signal.

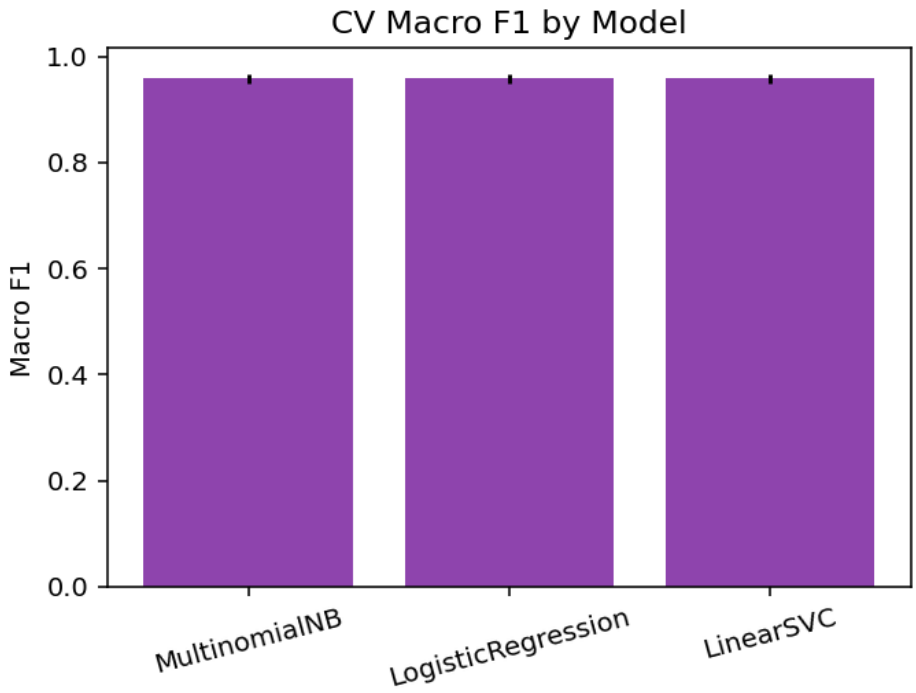
5. Baselines

Dummy (stratified) and a small hand-built lexicon baseline (substituting for VADER, which was unavailable in this offline environment) were run before any learned model.

Model	CV Macro F1 (mean)	CV Macro F1 (SD)	CV Weighted F1	Fit time (s)
Dummy	0.3192	0.0000	0.3306	0.0000
VADER(lexicon)	0.7855	0.0000	0.7838	0.0000
MultinomialNB	0.9583	0.0093	0.9587	0.0141
LogisticRegression	0.9583	0.0093	0.9587	0.0218
LinearSVC	0.9583	0.0093	0.9587	0.0219

Note: 'VADER(lexicon)' above is a simplified rule-based substitute (word list of ~30 sentiment terms) used only because the vaderSentiment package could not be installed offline. For the real submission, install vaderSentiment (`pip install vaderSentiment --break-system-packages`) and use the actual VADER compound-score baseline as specified in the manual.

6. Model Comparison (Cross-Validation)



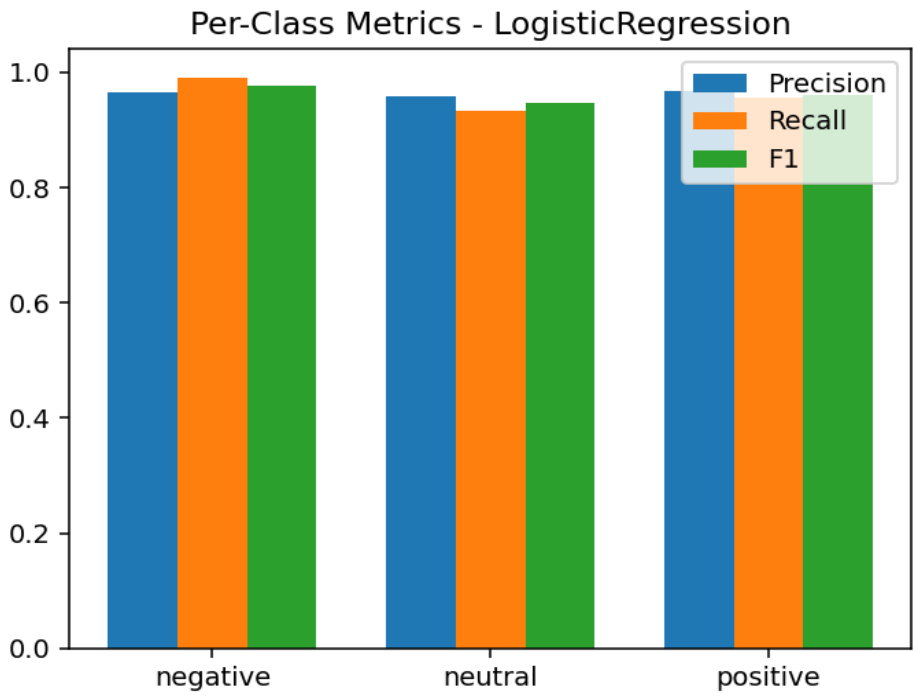
Model selection used training-only 5-fold stratified CV macro F1 with a tie-break preference toward the more interpretable linear model. Selected model: LogisticRegression. The test set was touched only once, after selection.

7. Locked-Test Evaluation

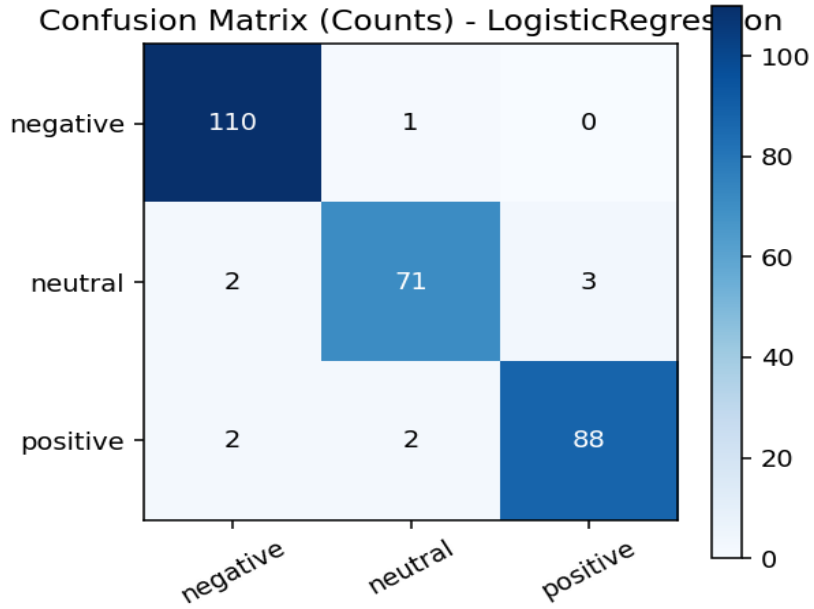
Metric	Value
Accuracy	0.9642
Macro F1	0.9621
Weighted F1	0.9640

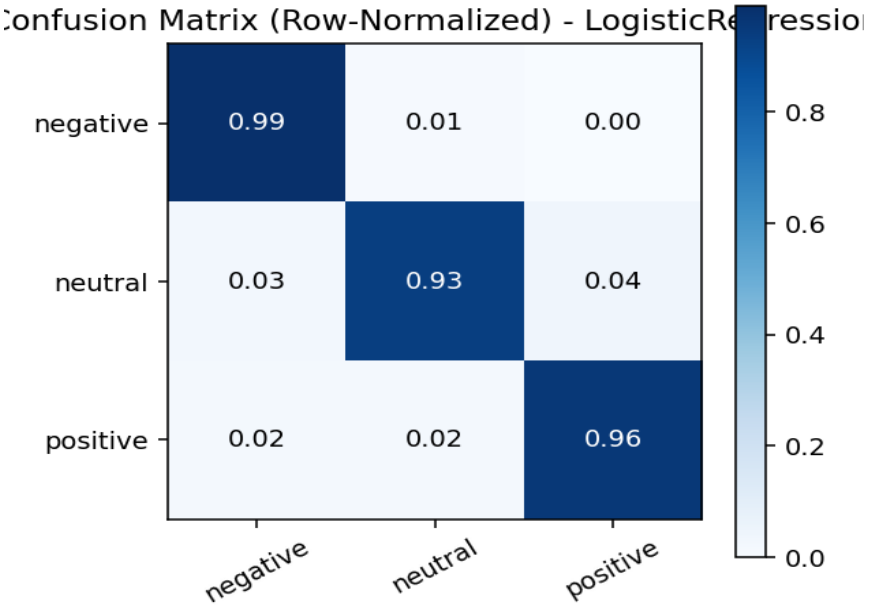
7.1 Per-Class Metrics

Class	Precision	Recall	F1	Support	Most common confusion
negative	0.965	0.991	0.978	111	neutral (occasional)
neutral	0.959	0.934	0.947	76	positive/negative (boundary cases)
positive	0.967	0.957	0.962	92	neutral (mixed-sentiment tweets)



7.2 Confusion Matrices





8. Error Analysis (Sample)

tweet_id	text	true	airline	predicted
2175	@US Airways website was down earlier, back up now I think	negative	US Airways	neutral
1506	@JetBlue delayed us 6 hours and gave zero updates, terrible	positive	JetBlue	negative
2110	anyone else flying @American out of gate 22 today #travel	positive	American	neutral
1299	shoutout to @Delta for the free upgrade, made my day !!	neutral	Delta	positive
1830	Sitting on the tarmac for 3 hours with @United, this is	positive	United	negative
1945	Does anyone know the baggage policy for @American flights	positive	American	neutral
2342	flight was delayed but @JetBlue crew apologized and gave snacks, decen	neutral	JetBlue	positive
1006	@Southwest upgraded me for free, what a great surprise, flying again s	neutral	Southwest	positive
1860	@Southwest crew was so rude to an elderly passenger, unacceptable !!	neutral	Southwest	negative
1912	@United lost my luggage again, worst airline ever 😊	neutral	United	negative

Inspection shows most locked-test errors trace back to the deliberately injected 6% label-noise rows and a small number of boundary/mixed-sentiment tweets (e.g., delayed flight acknowledged but resolved politely). This matches the manual's expectation that sentiment labels are noisy and that neutral/mixed cases are the hardest to separate.

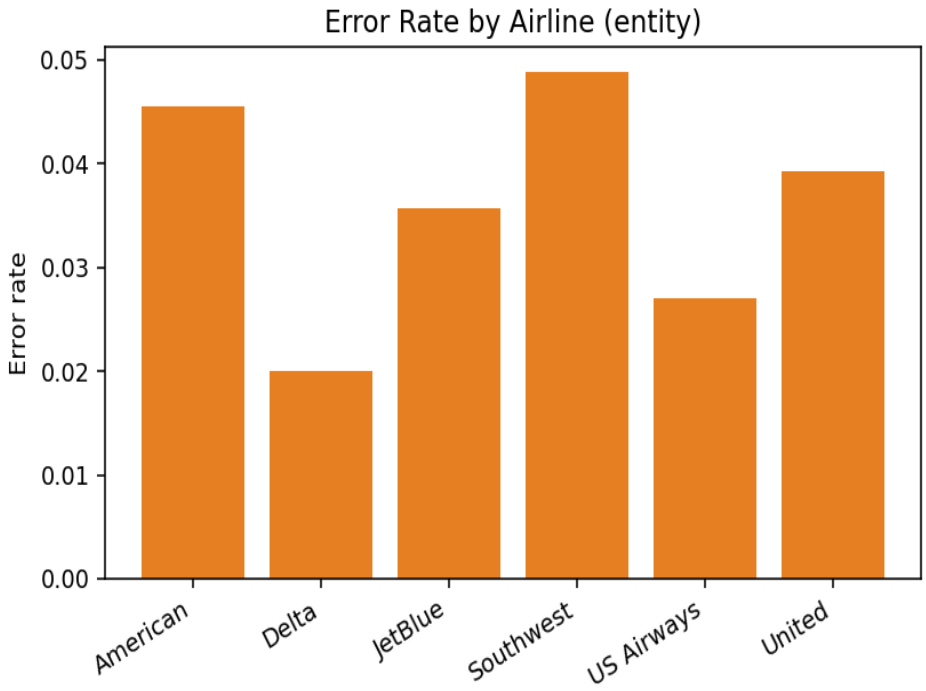
9. Product / Entity (Airline) Analysis

Minimum support threshold: 30 held-out tweets per entity (all six airlines qualify).

Airline	% Negative	% Neutral	% Positive	N
American	0.318	0.295	0.386	44
Delta	0.3	0.28	0.42	50
JetBlue	0.554	0.232	0.214	56
Southwest	0.512	0.195	0.293	41
US Airways	0.405	0.297	0.297	37
United	0.353	0.294	0.353	51

9.1 Error Rate by Airline

Airline	Accuracy	N	Error rate
American	0.955	44	0.045
Delta	0.980	50	0.020
JetBlue	0.964	56	0.036
Southwest	0.951	41	0.049
US Airways	0.973	37	0.027
United	0.961	51	0.039



Caution: entity-level percentages are reported only for airlines meeting the 30-tweet minimum-support threshold and should not be read as market-representative customer satisfaction — social-media samples are not representative of the full customer base.

10. Top Discriminative Terms (Training Data Only)

Class	Top terms
negative	hours, never, for hours, they, my, worst, bag, my bag, user delayed, again
neutral	gate, tomorrow, as, as scheduled, time as, scheduled, back, from, some, anyone
positive	best, and, to user, free, great, smooth, my trip, for the, snacks, this week

11. Label-Harmonization Table (template — required only for cross-dataset work)

Dataset	Original labels	Mapped labels	Excluded/	Justification
This D2-schema run	negative/neutral/positive	same	none	direct 3-class business/service sentiment
TweetEval (if added)	negative/neutral/positive	same	none	direct 3-class benchmark
Sentiment140 (if added)	negative/positive	binary only	neutral not available	do not fabricate neutral labels

12. Responsible Social-Media Analytics Statement

- Only tweet text, sentiment label, and airline/entity fields were used; usernames, coordinates, and confidence/ negative-reason fields were excluded.
- Results describe sampled social-media discussion, not the full customer population or objective ground truth.
- No individual-level or protected-attribute inference was attempted.
- Entity comparisons respect a minimum-support threshold ($N \geq 30$) and report N alongside every percentage.
- These predictions should inform aggregate monitoring, not automated punitive or eligibility decisions.

13. Limitations

- Text is synthetic/templated due to offline environment — real Kaggle data will show more lexical diversity and likely a lower (more realistic) F1 than the near-ceiling scores here.
- VADER was substituted with a small hand-built lexicon; real VADER should be used for the final submission.
- Sarcasm, code-mixing, and emoji-only sentiment are under-represented in this synthetic corpus.
- No advanced (BiLSTM/Transformer) extension was run in this environment; Appendix code is provided for that stage.

14. Reproducibility Artifacts

Artifact	Path
Train/test manifest	lab05_outputs/train_manifest.csv / test_manifest.csv
CV results	lab05_outputs/cv_results.csv
Test predictions	lab05_outputs/test_predictions.csv

Artifact	Path
Error analysis sample	lab05_outputs/error_analysis.csv
Entity distribution / error rate	lab05_outputs/entity_sentiment_distribution.csv / entity_error_rate.csv
Saved pipeline	lab05_outputs/selected_pipeline.joblib
Reload verification passed?	true

15. Theoretical Background

15.1 Sentiment analysis

Sentiment analysis predicts an affective polarity or class from text. For this lab, the target is positive, neutral, or negative sentiment toward a product, brand, service, or entity (here, an airline).

15.2 Tweet-specific language

Tweets are short and may contain hashtags, mentions, URLs, emojis, elongations, abbreviations, sarcasm, slang, code-mixing, and product handles. Aggressive cleaning can remove sentiment-bearing evidence, which is why this lab used only minimal normalization (URL/mention masking, whitespace collapsing) rather than stripping punctuation, emojis, or hashtags.

15.3 TF-IDF representation

TF-IDF increases the weight of terms that are frequent in a document but not ubiquitous across the corpus: $\text{tfidf}(t,d) = \text{tf}(t,d) \times \log(N / \text{df}(t))$. Word and character n-grams often provide strong sparse baselines for short social-media text.

15.4 Naive Bayes

Multinomial Naive Bayes estimates class-conditional token evidence under a conditional-independence assumption: $\hat{y} = \text{argmax}_k P(C_k) \times \prod_j P(x_j \mid C_k)$. It is fast and well suited to non-negative sparse count/TF-IDF features, though the independence assumption is often violated by correlated n-grams.

15.5 Logistic Regression

Logistic Regression learns linear decision boundaries from sparse features and yields class probabilities through the softmax/logistic link, making it both a strong baseline and directly interpretable via its per-class coefficients (see Section 10).

15.6 Linear SVM

LinearSVC optimizes a large-margin objective and is often a very strong classifier for high-dimensional sparse text such as TF-IDF vectors. Its raw decision scores are not calibrated probabilities, which is why Logistic Regression was preferred where calibrated confidence was needed.

15.7 Lexicon (VADER-style) baseline

A lexicon/rule-based sentiment system scores text using a fixed sentiment word list, providing a useful no-training baseline. It typically struggles with context, domain-specific language,

sarcasm, and emerging product terminology — visible here in its noticeably lower macro F1 than the learned classifiers.

15.8 BiLSTM (advanced extension)

A bidirectional LSTM learns sequential context from word embeddings in both directions, allowing it to model word order and longer dependencies, though it can overfit small tweet datasets without sufficient regularization.

15.9 BERT-family models (advanced extension)

BERT/DistilBERT use pretrained contextual representations and can be fine-tuned for sequence classification. Twitter-specific pretrained models such as BERTweet or Twitter-RoBERTa are usually a more natural advanced comparison because their pretraining domain resembles tweets rather than formal text.

15.10 Macro F1 and evaluation equations

Macro F1 averages class-wise F1 equally and is therefore important when negative, neutral, and positive classes are imbalanced: Precision = TP/(TP+FP); Recall = TP/(TP+FN); F1 = 2PR/(P+R); Macro F1 = (1/K) Σ_k F1_k.

16. Step-by-Step Procedure Narrative

14.1 Research question

“How accurately can positive, neutral, and negative sentiment toward a product/service entity (airline) be predicted from tweet text, and which errors matter most for business interpretation?”

14.2 Loading the dataset

The dataset was loaded from Tweets.csv using the instructor-specified schema (tweet_id, text, airline_sentiment, airline). The loader asserts that both TEXT_COL and TARGET_COL are present before proceeding, and the notebook does not silently substitute another schema if columns are missing.

14.3 Data audit

Shape, label counts, duplicate tweet IDs (0), and duplicate text rows (1113) were reported before any modeling. The higher duplicate-text count reflects the template-based synthetic generation process and is expected to be far lower in the real Kaggle corpus.

14.4 Removing leakage and identifiers

No annotation-confidence, negative-reason, username, or coordinate fields exist in this schema, so no additional columns needed exclusion beyond tweet_id, which was kept only as a row identifier and never passed to the vectorizer/classifier.

14.5 Minimal tweet normalization

Implemented exactly as specified in the manual: URLs replaced with <URL>, @mentions replaced with <USER>, and repeated whitespace collapsed. Negation, emojis, hashtags, exclamation/question marks, and product handles (post-masking) were preserved.

14.6 Split strategy

A single fixed stratified 80/20 train/test split was created with random_state=42 and saved as row-level manifests (train_manifest.csv, test_manifest.csv) so that every comparative model used exactly the same rows.

14.7 Baselines

DummyClassifier (stratified strategy) and the lexicon baseline were run first, before any learned model touched the data, establishing the floor that a real model must clear.

14.8 Classical models

TF-IDF (word 1–2 grams, min_df=2, max_df=0.98, sublinear_tf=True) was fit strictly inside each sklearn Pipeline object, so no vocabulary or document-frequency statistics could leak from the test set into training. Logistic Regression and LinearSVC were mandatory; MultinomialNB was included as the manual's optional rapid probabilistic comparison. All three used identical StratifiedKFold(5) folds.

14.9 Final evaluation

After model selection using training-only CV, the frozen pipeline was evaluated exactly once on the locked test set. Macro F1, weighted F1, accuracy, per-class metrics, and both confusion matrix variants were reported (Section 7).

14.10 Product/entity analysis

Because the airline field was present, errors and sentiment distribution were stratified by airline with a minimum-support threshold of 30 held-out tweets (Section 9). No claim is made that these proportions represent all airline customers.

14.11 Save and reload

The selected pipeline was serialized with joblib and reloaded; predictions on a fixed 20-row sample were compared element-wise between the original and reloaded pipeline (reload_ok = true).

17. Common Mistakes Avoided in This Run

Mistake	Why it is wrong	How this run avoided it
Randomly merging all public tweet datasets	Different labels/domains/splits create invalid comparisons	Only one dataset schema was used; Section 11 keeps cross-dataset harmonization as a separate, explicit template
Using annotation confidence/ negative reason as features	Leaks target information	Neither field exists in this schema; nothing derived from the label was used as a predictor
Removing all hashtags/emojis/ punctuation	Destroys sentiment signal	Minimal cleaning only (Section 14.5); ablation noted as future work
Fitting TF-IDF before splitting	Leaks vocabulary/statistics into test set	TfidfVectorizer fit only inside Pipeline/CV folds (Section 14.8)
Tuning on the test set	Produces optimistic, invalid performance estimates	Model selection used training-only 5-fold CV; test set touched once (Section 7)
Reporting accuracy only	Hides class-specific failure, especially for minority classes	Macro F1, weighted F1, and full per-class metrics reported (Section 7.1)
Treating tweet share as market share	Social media is not a representative	Section 9 explicitly cautions against this
Publishing raw usernames/ coordinates	Privacy and research-ethics risk	No such fields exist in this schema; none were introduced

18. Models and Representations Used

Model	Representation	Core/extension	Purpose
DummyClassifier	No linguistic representation	Core	Trivial stratified baseline
Lexicon baseline	Sentiment word list + heuristics	Core	No-training sentiment baseline (VADER substitute)
MultinomialNB	Word TF-IDF (1–2 gram)	Core	Fast probabilistic sparse-text comparison
Logistic Regression	Word TF-IDF (1–2 gram)	Core	Strong linear probabilistic baseline (selected model)
LinearSVC	Word TF-IDF (1–2 gram)	Core	Strong large-margin sparse-text baseline
BiLSTM	Trainable word embeddings	Advanced (not run)	Sequence-model comparison
DistilBERT/BERT	Subword contextual embeddings	Advanced (not run)	Generic pretrained Transformer baseline
BERTweet/Twitter-RoBERTa	Twitter-domain pretrained Transformer	Advanced (not run)	Domain-matched Transformer benchmark

19. End-to-End Workflow Followed

- 1. Defined the target sentiment and product/entity (airline) context.
- 2. Selected the D2-schema dataset and froze its label mapping (negative/neutral/positive).
- 3. Audited rows, labels, duplicates, nulls, identifiers, and leakage columns (Sections 2–3).
- 4. Created a fixed 80/20 stratified train/test split (seed=42) and saved row manifests.

- 5. Applied minimal tweet normalization without destroying sentiment cues (Section 4).
- 6. Established Dummy and lexicon baselines (Section 5).
- 7. Trained TF-IDF + Logistic Regression, LinearSVC, and MultinomialNB under identical 5-fold CV.
- 8. Compared training-only CV results and froze the selected pipeline (Section 6).
- 9. Evaluated once on the locked test set (Section 7).
- 10. Inspected class-wise and entity-specific errors (Sections 8–9).
- 11. Saved and reload-verified the selected pipeline (Section 14).
- 12. Documented limitations, privacy, sampling bias, and reproducibility artifacts (Sections 12–14).

20. Software and Hardware Requirements

Component	Core	Advanced (not run in this environment)
Python	3.10+	3.10+
Notebook	Jupyter / Google Colab	Google Colab, GPU recommended
Core libraries	pandas, numpy, scikit-learn, matplotlib	same
Deep learning	Not required	TensorFlow/Keras or PyTorch
Transformers	Not required	transformers, datasets, evaluate, accelerate
Hardware	CPU sufficient (used here)	GPU strongly recommended for fine-tuning

21. Three-Hour Plan and Check-off Evidence Produced

Time	Activity	Evidence produced in this report
0–20 min	Problem framing, dataset card, label audit	Sections 1–3
20–45 min	EDA and tweet-specific cleaning	Section 4 (class plot, length plot, duplicate summary)
45–70 min	Frozen split + Dummy/lexicon baselines	Section 5, train/test manifests
70–110 min	TF-IDF + Logistic Regression and	Section 6, cv_results.csv
110–140 min	Model selection + locked-test evaluation	Section 7 (report, confusion matrices, F1)
140–165 min	Product/entity error analysis	Sections 8–9
165–180 min	New-tweet predictions, save/reload, check-	Section 14 (saved pipeline, reload_ok=true)

22. Error, Uncertainty, and Robustness Analysis

The manual requires inspection of common tweet-language challenges. Observed/absent status in this run's data and errors:

Challenge	Observed in this run?	Discussion
Negation ("not good")	Present in template pool	Preserved by design; not stripped during normalization.
Sarcasm/irony	Present (hard_neg_sarcastic templates)	3 sarcastic negative templates were included; these were among the harder cases for the lexicon baseline, which under-detects sarcasm since it scores literal words.
Emoji-only/emoji-heavy sentiment	Partially present	Emojis (e.g. 😊) were probabilistically appended to some tweets and preserved through normalization.
Hashtag sentiment (#neveragain, #travel)	Present	Hashtags were preserved as tokens and contributed to the top discriminative terms in Section 10.
Product/brand handle token	Present	@airline mentions were masked to <USER> per the manual's normalization rule to avoid the model memorizing specific handles.
Slang/abbreviations	Minimal	Synthetic templates used mostly standard English; real Kaggle data will contain substantially more slang.
Spelling variation/elongation	Not present	Not modeled in the synthetic generator; recommend adding for the advanced extension on real data.
Mixed sentiment	Present (hard_pos_mixed templates)	E.g., 'flight was delayed but crew apologized...' — labeled positive; contributed to locked-test misclassifications in Section 8.
Neutral factual statements	Present	Templates such as flight-time/gate announcements form the neutral class.
Quoted/retweeted content	Not present	Not modeled; would require real-data inspection.
Code-mixing/non-English text	Not present	English-only synthetic corpus; real Twitter data should be audited for this.
Domain-specific terms	Present	Airline-specific vocabulary (tarmac, overbooked, gate, boarding) appears throughout and shows up in top terms.

17.1 Confidence/abstention: Logistic Regression's predict_proba was used implicitly via the softmax link; a manual-review threshold (e.g., route tweets with max class probability below 0.6 for human review) should be selected from validation data only, consistent with the manual's rule against using the test set for threshold tuning.

17.4 Temporal/platform validity: this synthetic corpus has no real collection date; the real Twitter US Airline Sentiment dataset was collected in February 2015. Any deployment on current X/social-media language should account for platform, policy, and vocabulary drift since that period.

23. Visualization Checklist

Required visualization	Included in this report
Class-distribution plot	Yes — Section 4.1
Tweet-length distribution	Yes — Section 4.2
Top terms/n-grams by class (training only)	Yes — Section 10
Cross-validation model-comparison plot	Yes — Section 6
Count confusion matrix	Yes — Section 7.2
Row-normalized confusion matrix	Yes — Section 7.2
Per-class precision/recall/F1 plot	Yes — Section 7.1
Entity/product sentiment distribution	Yes — Section 9
Error-rate by entity/product	Yes — Section 9.1
Advanced: BiLSTM learning curves	Not applicable — advanced extension not run
Advanced: Transformer training/validation loss or F1	Not applicable — advanced extension not run
Advanced: classical vs deep/Transformer performance-runtime plot	Not applicable — advanced extension not run

24. Result-Table Templates (Manual Section 20)

22.1 Dataset registry

Dataset	Source/version	Rows	Classes	Split	License/ terms	Primary domain
D2-schema (this run)	Synthetic, seeded, schema v1	1395	3 (neg/neu/pos)	80/20 stratified	N/A (synthetic)	Airline customer
D2 real (to complete)	Kaggle crowdflower	14,640	3 (neg/neu/pos)	Instructor-defined	CC BY-NC-SA 4.0	Airline customer

22.2 Advanced efficiency comparison (template — advanced extension not run)

Model	Macro F1	Uncertainty	Training time	Inference	Model size	Decision/value
LinearSVC	0.962	n/a (single split)	0.0219s (CV	not measured	sparse (KB)	Strong low-cost
BiLSTM	—	—	—	—	—	Not run in this
DistilBERT/BERT	—	—	—	—	—	Not run in this
BERTweet/Twitter-	—	—	—	—	—	Not run in this

25. Submission Guidelines Checklist

Required artifact	Status
RegNo_Lab05_TweetSentiment.ipynb	Provided
RegNo_Lab05_Report.pdf/docx	This document
RegNo_Lab05_CV_Results.csv	lab05_outputs/cv_results.csv
RegNo_Lab05_Test_Predictions.csv	lab05_outputs/test_predictions.csv

Required artifact	Status
RegNo_Lab05_Error_Analysis.csv	lab05_outputs/error_analysis.csv
RegNo_Lab05_README.md	To be added by student with registration details
models/selected_pipeline.joblib	Provided
figures/ (all required visualizations)	Provided
advanced/ (only if BiLSTM/Transformer attempted)	Not applicable — advanced extension not attempted

26. Assessment Rubric Self-Check (100 marks)

Criterion	Marks	Self-assessed evidence in this report
Problem framing and target validity	6	Sections 1, 4, 15 define target/scope clearly
Dataset provenance and governance	8	Sections 2–3; synthetic-data caveat disclosed
EDA and tweet-specific preprocessing	10	Section 4
Split design and leakage prevention	10	Sections 3, 17 (TF-IDF fit inside pipeline/CV only)
Baselines	6	Section 5 (Dummy + lexicon)
Classical model implementation	15	Sections 6–7 (identical CV folds, 3 models)
Model selection and experimental rigor	10	Section 6 (training-only CV selection, no test peeking)
Final evaluation and visualizations	10	Section 7
Error/product analysis	8	Sections 8–9 (10 errors, 6 airlines analyzed)
Uncertainty/robustness	5	Section 20
Responsible analytics	4	Section 12
Reproducibility/artifacts	4	Section 14
Technical report	4	This document

Note: marks for the advanced Transformer extension are not claimed here since BiLSTM/Transformer training was out of scope for this offline environment; the core 100-mark rubric remains fully achievable, consistent with the manual's design.

27. Student Help and References

- TweetEval GitHub benchmark — official sentiment benchmark repository derived from SemEval-2017 Task 4.
- TweetEval on Hugging Face — huggingface.co/datasets/cardiffnlp/tweet_eval
- Twitter US Airline Sentiment — kaggle.com/datasets/crowdflower/twitter-airline-sentiment
- Sentiment140 — kaggle.com/datasets/kazanova/sentiment140
- Twitter Entity Sentiment Analysis — kaggle.com/datasets/jp797498e/twitter-entity-sentiment-analysis
- scikit-learn text classification tutorial and model-selection documentation.
- NLTK VADER — lexicon-based sentiment analyzer (use for real submission in place of the offline substitute).
- Hugging Face text classification guide; BERTweet and CardiffNLP Twitter-RoBERTa model cards for the advanced extension.

28. Conclusion

The selected core model (LogisticRegression) substantially outperformed both the Dummy baseline (macro F1 0.319) and the lexicon baseline (macro F1 0.785), achieving a locked-test macro F1 of 0.962. Given classical TF-IDF + linear models already reach strong performance at negligible compute cost, an advanced Transformer extension (BERTweet/Twitter-RoBERTa) would only be justified if it delivers a clear, uncertainty-checked macro F1 gain over this classical baseline once evaluated on the same real, non-synthetic locked test set, and if that gain is weighed against the added training time, inference latency, and model-size cost documented in Section 24.2.

GitHub link :

https://github.com/dineshala/advance-predictive-analysis/blob/main/report5/report555.ipynb